

Layer 1: Schema Auto-Repair

JSON schema validation → LLM repair prompt → Retry (max 2)

↓ If schema valid

Layer 2: SCAD Auto-Fix

Detect hull/minkowski/high-\$fn → Simplify → Extended timeout

↓ If renders OK

Layer 3: VLM Correction

6-view render → GPT-5.2 assessment → LLM fix → Iterate (max 3)

↓ If VLM approves

Layer 4: Component Verification

Tiered part check → Connectivity check → Targeted fix (max 3)

↓ If parts present

Layer 5: Manual Repair

User hint (e.g., "make wheels bigger") → Incorporate into prompt